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Find the area of solid
revolution about x-axis)

1) $x \in [1, 5]$ $f(x) = \sqrt{x+1}$

2) $x \in [\frac{\pi}{2}; \pi]$ $f(x) = \sin x$

$$S = \int_a^b 2\pi f(x) \sqrt{1 + (f'(x))^2} dx$$

✓3

Find the arc length

1) $x \in [2, 3]$ $y \geq 0$ $36y^2 = (x^2 - 4)^3$

2) $x \in [1, 3]$ $f(x) = \frac{x^3}{3} + \frac{1}{4}x$

$$L = \int_a^b \sqrt{1 + (f'(x))^2} dx$$

$\forall p(x) \deg p(x) \geq 3 \Rightarrow \exists p_1, \dots, p_n \in \mathbb{R}[x] \text{ s.t. } \deg p_i \leq 2 \quad p = p_1 \dots p_n$

$$(p_1(x+1) + p_2(x+1)) = 4x$$

$\sqrt{1}$

$$a) \int \frac{4x^2}{x^4 + x^3 + x^2 + x} dx = \int \frac{4x}{x^3 + x^2 + x + 1} = \int \frac{4x}{(x+1)(x^2+1)} = \int \frac{q_1}{x+1} + \int \frac{q_2}{x^2+1} = \int -\frac{2}{x+1} + \frac{2x+2}{x^2+1} dx$$

Check each integral is convergent or divergent. If convergent, evaluate it.

$$a) \int_{-\infty}^0 \frac{1}{3-4x} dx$$

$$b) \int_0^{\infty} \frac{1}{\sqrt{x} + x\sqrt{x}} dx$$

$$b) \int_0^1 \ln^2 x dx$$

$\lim_{x \rightarrow 0} x \ln^2 x = 0$

$$c) \int_1^{+\infty} \frac{-x^2}{e^x} dx$$

Check whether the following integral convergent (composition test)

$$c) \int_1^{+\infty} \frac{\ln x |\cos^{3/5} x|}{x^2 + \frac{1}{x^2}} dx$$

$$d) \int_{-\pi}^{\pi} \frac{|\sin x|}{x^3 + \frac{4}{5} x^{3/2}} dx$$